

Iterative Self-Refinement for Multi-Camera Calibration via Learned Pose Conditioning

Abstract

We present a method for improving multi-camera extrinsic calibration using iterative self-conditioning with a learned geometry model. Starting from Pi3X, a permutation-equivariant feed-forward network that estimates camera poses from images, we observe that feeding its own predictions back as conditioning improves rotation estimates but degrades translations due to Sim(3) scale ambiguity. We propose a hybrid fusion strategy that decouples rotation and translation channels: rotation edges are extracted from conditioned inference while translation edges are preserved from unconditioned inference. Combined with Monte Carlo noise perturbation, this achieves 28% rotation improvement and 21% relative pose improvement over the baseline, with no external calibration data required.

1 Introduction

Extrinsic calibration of wide-baseline multi-camera systems is a prerequisite for 3D reconstruction, tracking, and scene understanding. Classical approaches rely on either physical calibration targets (checkerboards, ArUco markers) or structure-from-motion pipelines that extract and match distinctive static features across views. Both fail in a growing class of real-world deployments where: (i) cameras are spaced far apart with minimal field-of-view overlap, (ii) the dominant visual content is dynamic and repetitive, and (iii) the static scene geometry is degenerate.

We study this problem in a concrete setting—eight wide-baseline cameras monitoring a cattle feedlot—that instantiates all three failure modes simultaneously:

- **Wide baseline, minimal overlap.** Cameras are separated by 5–10m with narrow overlapping regions, making cross-view feature matching unreliable.
- **Dynamic, repetitive content.** Each frame is occupied by moving cattle with near-identical black appearance. Matched features across views may correspond to different animals.
- **Degenerate static structure.** The pen is a flat ground plane with few distinctive static landmarks—violating the non-planar scene assumption required for robust triangulation.

These conditions are not unique to livestock monitoring. The same challenge arises in wildlife observation (savanna, marine), sports analytics (stadium cameras observing players), traffic monitoring at wide intersections, and construction site surveillance. The common thread is *calibrating cameras whose observations are dominated by dynamic, visually repetitive content over geometrically degenerate backgrounds*.

1.1 Foundation Geometry Models as a Solution

Recent feed-forward geometry models [15, 13, 14] bypass explicit feature correspondence by learning to reason about multi-view geometry directly from images. Pi3X [15], a permutation-equivariant model that processes all N views jointly, achieves state-of-the-art accuracy on standard benchmarks. Applied to our setting, Pi3X succeeds where classical methods fail entirely—producing camera poses with $\sim 4.4^\circ$ mean rotation error from raw images alone.

However, this accuracy saturates. Standard geometric post-processing (Sim(3) gauge fixing, robust rotation averaging, pose graph optimization) barely improves upon Pi3X’s raw output, because the residual error is not statistical noise but systematic bias arising from the scene’s weak geometric signal.

1.2 Key Insight: Self-Conditioning on Sim(3)

Pi3X accepts optional pose conditioning—feeding known camera poses alongside images improves its predictions. We exploit this by feeding Pi3X’s *own output* back as conditioning, creating an iterative self-refinement loop analogous to fixed-point iteration on the pose manifold.

We discover a surprising asymmetry: **self-conditioning improves rotation but degrades translation**. This is not a bug but a fundamental consequence of Sim(3) ambiguity—Pi3X’s outputs are defined up to an arbitrary scale, so translation information fed back as SE(3) conditioning introduces compounding scale bias, while rotation (which is scale-invariant) contracts toward the correct solution.

This raises two questions that we address in this paper: *Under what conditions does self-conditioning converge on the rotation manifold?* And: *How can the rotation-translation asymmetry be exploited to improve calibration without degrading either component?*

1.3 Contributions

1. **Empirical characterisation of self-conditioning dynamics.** We show that learned geometry models act as contraction mappings on SO(3) (with measurable rate $\lambda \approx 0.8$) but *not* on the translation component of Sim(3). This is the first characterisation of convergence behaviour in self-conditioned feed-forward geometry models.
2. **Decoupled hybrid refinement.** We propose a method that exploits the R/t asymmetry by fusing rotation edges from conditioned inference with translation edges from unconditioned inference in a single pose graph. This preserves translation accuracy while capturing rotation improvements.
3. **Monte Carlo iterative denoising.** We introduce stochastic perturbation of the conditioning signal, interpreting conditioned inference as approximate posterior sampling. Combined with iterative hybrid fusion, this achieves 28% rotation improvement and 21% relative pose improvement over the baseline—with no external calibration data.
4. **Convergence analysis.** We predict the method’s accuracy floor from the measured contraction rate and validate experimentally, establishing when self-refinement helps and when it saturates.

Our method is applicable to any learned geometry model with a conditioning interface (Pi3X, VGGT, DUST3R variants) deployed in feature-adversarial environments where classical calibration is infeasible.

2 Related Work

Learned multi-view geometry. Feed-forward models for multi-view 3D reconstruction have advanced rapidly. DUS_t3R [14] introduced pairwise pointmap regression; MAS_t3R [6] extended it with local feature matching. Pi3X [15] and VGGT [13] process all N views jointly via permutation-equivariant architectures, achieving stronger global consistency. These models output camera poses and dense geometry from images alone, but their accuracy on wide-baseline, feature-adversarial scenes remains limited.

Multi-camera calibration. Classical extrinsic calibration uses physical targets (checkerboards [17]), vanishing points [3], or structure-from-motion pipelines [8]. Target-free methods assume sufficient static texture for feature matching across views—an assumption violated in dynamic, textureless environments. Our setting falls outside the scope of these methods.

Pose graph optimization. Rotation averaging [5, 2] and translation averaging [7] are classical tools for aggregating noisy pairwise relative poses into a globally consistent estimate. We adopt these as our post-processing backbone (Stage 1) and as the fusion mechanism for combining heterogeneous edge sources (Stages 3–4).

Test-time refinement. Test-time training (TTT) [10] and test-time adaptation adapt model predictions at inference time using self-supervised objectives. In geometry estimation, self-supervised depth refinement [4] fine-tunes networks on test sequences using photometric consistency. Our approach differs in that we do *not* modify model weights—instead, we exploit the model’s conditioning interface to iteratively refine predictions, treating the pretrained model as a fixed denoiser.

Iterative pose refinement. Classical approaches such as ICP [1] and PnP+RANSAC refinement iterate between correspondence estimation and pose optimization. Bundle adjustment [11] jointly optimises poses and 3D structure. Our method is conceptually similar—iterative refinement toward a fixed point—but operates through a learned model’s conditioning mechanism rather than explicit geometric optimization.

Denoising and diffusion on manifolds. Score-based diffusion models [9] learn to denoise data via iterative refinement from noise toward the data manifold. Recent work has extended diffusion to SE(3) [16] for molecular and pose generation. Our Monte Carlo perturbation strategy (Section 4.4) shares this structure: perturb the current estimate, apply a learned denoiser (Pi3X conditioning), and average. However, we operate in a discriminative setting (fixed pretrained model, no diffusion training) and exploit the observation that convergence is restricted to the SO(3) component.

3 Foundation Geometry Model Benchmark

To select the best backbone for our calibration pipeline, we benchmark seven recent foundation geometry models on our dataset. All models receive the same 8 synchronised camera images per timestep and must output camera poses without any prior information.

3.1 Experimental Protocol

We sample 91 timesteps at regular intervals from synchronised video (2688×1520 resolution, resized per model requirements). For each model, we:

Table 1: Foundation geometry model benchmark on our wide-baseline, feature-adversarial dataset. Models are grouped by type (N-view joint vs. pairwise with post-hoc alignment) and ranked by ATE within each group. “Cond.” indicates whether the model accepts pose/intrinsic priors as conditioning input—a prerequisite for our self-refinement method. All metrics use Sim(3)-aligned predictions; C2C3 measures co-located camera consistency. Pairwise models use only 20 timesteps due to computational cost.

Model	Venue	Type	Cond.	ATE↓	ARE↓	RPE _R ↓	RPE _t ↓	C2C3↓	Time (s)
<i>N-view (joint)</i>									
Pi3X	CVPR’25	N-view	✓	0.177	4.37°	6.26°	0.39	0.088	106
MapAnything	CVPR’25	N-view	—	0.627	9.17°	14.19°	0.93	0.313	86
VGGT	CVPR’25	N-view	✓	1.248	10.48°	12.82°	1.55	2.011	75
NoPo4D	arXiv’25	N-view	—	2.556	62.06°	77.01°	3.17	0.627	214
Fast3R	CVPR’25	N-view	—	2.743	96.64°	102.88°	4.36	3.183	460
<i>Pairwise (post-hoc alignment)</i>									
Pow3R	CVPR’25	Pairwise	—	1.835	180°	28.53°	7.24	0.218	497
MonST3R	ICLR’25	Pairwise	—	2.607	47.24°	23.87°	3.56	0.111	1492

1. Run inference on all timesteps (8 cameras per timestep)
2. Aggregate per-timestep poses via quaternion averaging (rotations) and median (translations)
3. Apply Sim(3) Procrustes alignment to ground truth (with reflection check)
4. Compute ATE, ARE, RPE, and co-located pair distances

Ground truth poses are obtained from COLMAP applied to a static scene capture (no cattle present). Two camera pairs (C2–C3 and C5–C6) share physical mounting poles, providing a structural validation metric (ground truth distance ≈ 0).

3.2 Results

3.3 Analysis

The benchmark reveals a clear hierarchy:

Tier 1: Pi3X. Achieves sub-5° rotation error and 0.177 ATE—3.5× better than the next model. Its permutation-equivariant architecture that processes all 8 views jointly is well-suited to wide-baseline multi-camera geometry.

Tier 2: MapAnything, VGGT. Both achieve $\sim 10^\circ$ rotation error. Usable but significantly less accurate than Pi3X on this task.

Tier 3: Remaining models. MonST3R, Pow3R, NoPo4D, and Fast3R all exceed 20° rotation error—effectively failing on this dataset. The pairwise models (MonST3R, Pow3R) struggle because their post-hoc global alignment amplifies errors when inter-camera overlap is minimal.

Key observations.

- N-view joint models (Pi3X, MapAnything, VGGT) consistently outperform pairwise models, confirming that joint multi-view reasoning is essential for wide-baseline configurations.
- The co-location metric (C2C3) discriminates structural understanding: Pi3X correctly identifies that C2 and C3 share a pole (0.088), while VGGT (2.011) and Fast3R (3.183) place them far apart.
- All models were published in 2025, indicating that wide-baseline feature-adversarial calibration remains a frontier challenge for even the latest foundation models.

Based on this benchmark, we select Pi3X as our backbone and develop our self-refinement method (Sections 4.2–4.4) to push beyond its $\sim 4.4^\circ$ accuracy ceiling.

4 Method

Our method operates in four stages. Stage 1 applies standard geometric post-processing to establish a baseline estimate. Stages 2–4 constitute our contribution: self-conditioned iterative refinement with decoupled rotation-translation fusion.

4.1 Preliminaries: Geometric Post-Processing (Stage 1)

Given $N_t = 91$ timesteps of N -camera images, we run Pi3X to obtain per-timestep poses $\{T_k^c\}_{k=1}^{N_t}$ for each camera $c \in \{1, \dots, N\}$, where each $T_k^c \in \text{SE}(3)$ is defined up to a per-timestep $\text{Sim}(3)$ ambiguity (i.e., each timestep’s output coordinate frame is arbitrary in position, orientation, and scale). We aggregate these into a single estimate via three standard steps.

Gauge fixing. Since each timestep uses an arbitrarily chosen global frame, we align all timesteps to a reference via $\text{Sim}(3)$ Procrustes [?]:

$$\{s^*, R^*, \mathbf{t}^*\}_{=s,R,\mathbf{t}} \sum_{c=1}^N \|sR\mathbf{p}_k^c + \mathbf{t} - \mathbf{p}_{\text{ref}}^c\|^2 \quad (1)$$

where $\mathbf{p}_k^c \in \mathbb{R}^3$ denotes the position of camera c at timestep k , and $s \in \mathbb{R}^+$, $R \in \text{SO}(3)$, $\mathbf{t} \in \mathbb{R}^3$ parametrise the $\text{Sim}(3)$ transform.

Outlier rejection. For each camera, we project its N_t estimated positions onto the dominant 2D plane (via PCA on camera centres) and reject outliers whose in-plane distance exceeds $3 \times$ the Median Absolute Deviation (MAD). Inlier rotations are aggregated via the Fréchet mean on $\text{SO}(3)$ [?] $\text{—the rotation minimising the sum of squared geodesic distances to all inlier samples.}$

Pose graph optimization. We extract pairwise relative poses $T_{ij} = (T^i)^{-1}T^j$ from inlier timesteps, computing the Fréchet mean rotation and mean translation for each camera pair (i, j) . These relative poses serve as edges in a pose graph, weighted by the inverse variance of the per-timestep estimates (edges with higher consistency receive more weight). We then optimise globally via L_1 Weiszfeld rotation averaging on $\text{SO}(3)$ [5] followed by translation averaging with hard co-location constraints for cameras sharing a physical mounting pole.

Motivation for learned refinement. This pipeline reduces ATE by 10% ($0.177 \rightarrow 0.161$) but leaves ARE essentially unchanged ($4.37^\circ \rightarrow 4.39^\circ$). The residual rotation error is systematic—not reducible by further classical aggregation—because the scene provides weak geometric signal (flat pen, repetitive dynamic content). This motivates exploiting Pi3X’s conditioning interface to inject prior information.

4.2 Self-Conditioned Inference (Stage 2)

Pi3X supports multimodal conditioning: given approximate poses $\hat{T}^c$ and camera intrinsic matrices $\mathbf{\Pi}^c \in \mathbb{R}^{3 \times 3}$, it refines its predictions by attending to the provided geometric priors during encoding. We condition on the Stage 1 output:

$$\{T_k^{c,(1)}\} = f_\theta(\text{imgs}_k \mid \hat{T}^c, \mathbf{\Pi}^c), \quad k = 1, \dots, N_t \quad (2)$$

where f_θ denotes Pi3X and $\hat{T}^c$ is the Stage 1 estimate converted to the model’s input camera ordering.

Applying the same geometric post-processing to $\{T_k^{c,(1)}\}$ yields a refined estimate. We observe:

- ARE improves: $4.39^\circ \rightarrow 3.49^\circ$ (20% reduction)
- ATE degrades: $0.161 \rightarrow 0.166$ (3% increase)

Iterating naïvely amplifies both effects—rotation continues to improve (to $\sim 3.3^\circ$) while translation degrades monotonically (to 0.195 after 5 iterations). Section 5 provides a theoretical explanation: the rotation component contracts toward a fixed point on $\text{SO}(3)$, while translations diverge due to $\text{Sim}(3)$ scale mismatch.

4.3 Decoupled Hybrid Fusion (Stage 3)

Since conditioned inference improves rotation but degrades translation, we decouple them in the pose graph by sourcing edges from different inference runs. Let R_{ij} and $\mathbf{t}_{ij}$ denote the relative rotation and translation from camera i to camera j , computed as the mean over inlier timesteps. We construct fused edges:

$$\tilde{T}_{ij}^{\text{fused}} = \begin{pmatrix} R_{ij}^{\text{cond}} & \mathbf{t}_{ij}^{\text{uncond}} \\ \mathbf{0}^\top & 1 \end{pmatrix} \quad (3)$$

where R_{ij}^{cond} is the mean relative rotation extracted from conditioned inference (Stage 2) and $\mathbf{t}_{ij}^{\text{uncond}}$ is the mean relative translation from the original unconditioned run (Stage 1). Edge weights w_{ij} (inverse variance of the per-timestep relative pose estimates) are taken as $w_{ij} = \max(w_{ij}^{\text{cond}}, w_{ij}^{\text{uncond}})$.

The initial pose for graph optimization similarly uses conditioned rotations with unconditioned translations:

$$T_{\text{init}}^c = \begin{pmatrix} R_{\text{cond}}^c & \mathbf{t}_{\text{uncond}}^c \\ \mathbf{0}^\top & 1 \end{pmatrix} \quad (4)$$

This yields $\text{ARE} = 3.45^\circ$ (rotation improved) with $\text{ATE} = 0.159$ (translation preserved)—capturing the benefits of both sources.

Algorithm 1 Iterated Hybrid MC Refinement

Require: Pipeline estimate $\hat{T}^c$, images, intrinsics $\mathbf{\Pi}^c$

Require: Noise: σ_R, σ_t ; samples: S ; iterations: M

```
1:  $T_{\text{curr}} \leftarrow \hat{T}^c$ 
2: Compute unconditioned edges  $\{\mathbf{t}_{ij}^{\text{uncond}}, w_{ij}^t\}$  {fixed throughout}
3: for  $m = 1$  to  $M$  do
4:   for  $s = 1$  to  $S$  do
5:      $T_s^c \leftarrow \text{Perturb}(T_{\text{curr}}^c, \sigma_R, \sigma_t)$  for all  $c$ 
6:      $\{T_{k,s}^c\} \leftarrow f_\theta(\text{imgs} \mid T_s, \mathbf{\Pi}^c)$  for all  $k$ 
7:      $\hat{T}_s^c \leftarrow \text{Pipeline}(\{T_{k,s}^c\})$ 
8:   end for
9:   Extract rotation edges  $\{R_{ij}^{\text{cond}}, w_{ij}^R\}$  from pooled aligned poses
10:  Fuse:  $\tilde{T}_{ij} = (R_{ij}^{\text{cond}}, \mathbf{t}_{ij}^{\text{uncond}})$ 
11:   $T_{\text{opt}} \leftarrow \text{PoseGraph}(\tilde{T}_{\text{init}}, \tilde{T}_{ij}, w_{ij})$ 
12:   $T_{\text{curr}} \leftarrow (R_{\text{opt}}, \mathbf{t}_{\text{uncond}})$  {anchor translations}
13: end for
14: return  $T_{\text{opt}}$ 
```

4.4 Monte Carlo Iterative Refinement (Stage 4)

To further improve rotation estimates, we iterate the hybrid fusion with stochastic perturbation of the conditioning signal. The perturbation serves two purposes: (i) it explores the local neighbourhood of the current estimate, sampling multiple poses from the model’s “denoising basin”; (ii) it provides diversity across ensemble members, reducing the variance of the averaged rotation edges.

The Perturb operation applies independent Gaussian noise to each camera’s pose: rotation is perturbed via the exponential map $R' = R \cdot \exp([\epsilon_R]_\times)$ where $\epsilon_R \sim \mathcal{N}(\mathbf{0}, \sigma_R^2 I_3)$, and translation is perturbed additively $\mathbf{t}' = \mathbf{t} + \epsilon_t$ where $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \sigma_t^2 I_3)$.

Design choices.

- **Noise magnitude:** $\sigma_R = 2^\circ$ (axis-angle standard deviation), $\sigma_t = 0.05$ (isotropic, in scene units). These are chosen to stay within the model’s contraction basin ($< 5^\circ$; see Section 7).
- **Translation anchoring:** Translations are *never* updated from conditioned outputs—only from the original unconditioned baseline. This is the key invariant that prevents Sim(3) drift.
- **Early stopping:** Terminate when ARE improvement between iterations falls below 0.1° , or after $M = 3$ iterations (empirically sufficient; see Section 7).

Probabilistic interpretation. Each perturbed conditioned inference produces an approximate sample from the posterior $p(R \mid \text{imgs})$. Pooling $S \times N_t$ aligned poses across samples before extracting rotation edges estimates the posterior mean over relative rotations, reducing variance by $O(1/\sqrt{S})$. We formalise this connection to denoising score matching in Section 5.

5 Theoretical Analysis

We provide a formal interpretation of the self-conditioning dynamics observed in Section 4, explaining why rotation converges while translation diverges, and connecting the Monte Carlo perturbation

strategy to established frameworks.

5.1 Contraction Mapping on $\text{SO}(3)$

Let $f : \text{SO}(3)^N \rightarrow \text{SO}(3)^N$ denote the composite mapping from conditioning rotations to Pi3X’s output rotations after pipeline processing (gauge fix, outlier rejection, Fréchet mean). We define the geodesic distance on $\text{SO}(3)$ as:

$$d_{\text{SO}(3)}(R_1, R_2) = \|\log(R_1^\top R_2)\|_F \quad (5)$$

where $\log : \text{SO}(3) \rightarrow \mathfrak{so}(3)$ is the matrix logarithm. This equals the angle of the relative rotation $R_1^\top R_2$.

If f is a contraction mapping, i.e., there exists $\lambda < 1$ such that for all R in a neighbourhood of the true rotations R^* :

$$d_{\text{SO}(3)}(f(R), R^*) \leq \lambda \cdot d_{\text{SO}(3)}(R, R^*) \quad (6)$$

then by the Banach fixed-point theorem, the iteration $R^{(n+1)} = f(R^{(n)})$ converges to a unique fixed point R^∞ satisfying $f(R^\infty) = R^\infty$.

Empirical measurement. We measure the contraction rate by feeding GT poses corrupted with known noise levels and comparing input error to output error (Section 7, Table 4):

- Input error $4.4^\circ \rightarrow$ output error 3.5° ($\lambda \approx 0.80$)
- Input error $5.0^\circ \rightarrow$ output error 3.3° ($\lambda \approx 0.67$)
- Input error $2.0^\circ \rightarrow$ output error 1.6° ($\lambda \approx 0.80$)

The contraction rate is approximately $\lambda \approx 0.8$ in the $2\text{--}5^\circ$ regime. This predicts convergence to a fixed point at:

$$\text{ARE}^* = \frac{\lambda}{1 - \lambda} \cdot \epsilon_{\text{irr}} \approx 4 \cdot \epsilon_{\text{irr}} \quad (7)$$

where ϵ_{irr} is the model’s irreducible error (the residual even with perfect conditioning). With $\text{ARE}^* \approx 3.1^\circ$ observed experimentally, this gives $\epsilon_{\text{irr}} \approx 0.8^\circ$, consistent with the 1.2° achieved under perfect GT conditioning (which includes additional pipeline processing noise).

Basin of attraction. The contraction property holds only within a finite neighbourhood of R^* . Our noise robustness experiments (Section 7) show that conditioning with $> 10^\circ$ error causes catastrophic failure (output error jumps to 180°). This defines a basin of attraction of approximately $\pm 10^\circ$ around the true solution—our initial estimate (4.4° error) lies safely within it.

5.2 Failure of Contraction for Translation

Pi3X outputs poses in $\text{Sim}(3)$, not $\text{SE}(3)$. The translation component $\mathbf{t}^c$ is meaningful only relative to the per-run scale s . When fed back as $\text{SE}(3)$ conditioning, a scale mismatch accumulates:

$$\mathbf{t}^{(n+1)} = s^{(n)} \cdot g(\mathbf{t}^{(n)}, R^{(n)}) + \beta \quad (8)$$

where $s^{(n)} \neq 1$ in general and β represents systematic bias from the conditioning mechanism. Since the scale factor $s^{(n)}$ varies per iteration and is not corrected, this does not satisfy contraction:

$$\|\mathbf{t}^{(n+1)} - \mathbf{t}^*\| \not\leq \lambda \|\mathbf{t}^{(n)} - \mathbf{t}^*\| \quad (9)$$

This analysis motivates our hybrid fusion strategy: rotation contracts and should be iterated; translation does not and should be anchored to the unconditioned estimate.

5.3 Monte Carlo Perturbation as Posterior Sampling

The stochastic perturbation in Stage 4 admits an interpretation through the lens of denoising score matching [12]. Under the hypothesis that Pi3X’s conditioning mechanism was trained with noisy pose inputs (a common data augmentation for robustness), conditioned inference approximates the posterior mean:

$$f_{\theta}(R + \epsilon) \approx \mathbb{E}[R^* \mid R + \epsilon, \text{imgs}] \quad (10)$$

By running S perturbations $\{R \cdot \exp([\epsilon_s]_{\times})\}_{s=1}^S$ and computing the Fréchet mean of the outputs:

$$\bar{R} = \text{FréchetMean}(\{f_{\theta}(R \cdot \exp([\epsilon_s]_{\times}))\}_{s=1}^S) \quad (11)$$

we approximate the marginal posterior mean $\mathbb{E}[R^* \mid \text{imgs}]$ via Monte Carlo integration over the perturbation distribution. The variance of this estimator decreases as $O(1/\sqrt{S})$.

Remark. This interpretation relies on the assumption that Pi3X was trained with noisy conditioning. While we cannot verify this from the published model, the empirical contraction behaviour (smooth error reduction rather than mode collapse) is consistent with a model that has learned to denoise its conditioning inputs. The interpretation should be understood as a motivating analogy rather than a proven equivalence.

5.4 Connection to Iterative Denoising

The overall procedure—starting from a noisy estimate, applying a learned map that moves closer to the truth, and iterating—shares the structure of denoising diffusion [9]:

1. Start with a noisy estimate $R^{(0)}$ (unconditioned Pi3X output, $\sim 4.4^\circ$ error)
2. Apply a learned denoiser f_{θ} conditioned on images
3. The denoiser contracts the estimate toward the clean manifold
4. Iterate 2–3 times until convergence

The key differences from standard diffusion are: (i) we start close to the solution rather than from pure noise, requiring only 2–3 steps; (ii) we do not train a noise-level-conditional model; and (iii) convergence is restricted to the rotation subgroup, necessitating the hybrid decoupling. This suggests a broader principle: *pretrained geometry models with conditioning interfaces can serve as implicit denoisers at test time, even without explicit denoising training.*

6 Results

6.1 Experimental Setup

Dataset. We evaluate on 8 static cameras (2688×1520 resolution) deployed in a cattle feedlot. We sample $N_t = 91$ timesteps at regular intervals from synchronised video. Ground truth poses are obtained from COLMAP applied to a static capture of the same scene (without cattle).

Table 2: Progressive improvement from raw Pi3X through our full method. Cost is relative to a single unconditioned inference pass (~ 100 s on an A100). Row 3 shows naïve self-conditioning (rotation improves but translation degrades). Row 4 introduces hybrid fusion (translation preserved). Row 5 adds MC perturbation and iteration.

Method	ATE $\downarrow$	ARE $\downarrow$	RPE $_R\downarrow$	RPE $_t\downarrow$	C2C3 $\downarrow$	Cost
1. Raw Pi3X (quat average)	0.177	4.37°	6.26°	0.395	0.088	1×
2. + Pipeline (Stage 1)	0.161	4.39°	6.26°	0.375	0.000	1×
3. + Self-Conditioning (Stage 2)	0.166	3.49°	4.93°	0.384	0.000	2×
4. + Hybrid Fusion (Stage 3)	0.159	3.45°	4.93°	0.322	0.000	2×
5. + MC Iteration (Stage 4)	0.160	3.13°	4.57°	0.311	0.000	13×
<i>Oracle (GT pose conditioning)</i>	<i>0.097</i>	<i>1.22°</i>	—	—	—	1×

Evaluation metrics. All methods are evaluated after Sim(3) Procrustes alignment to ground truth (with reflection check):

- **ATE:** Absolute Translation Error (RMSE of camera position errors)
- **ARE:** Absolute Rotation Error (mean geodesic distance to GT rotations, in degrees)
- **RPE rot / trans:** Relative Pose Error averaged over all $\binom{8}{2} = 28$ camera pairs
- **C2C3:** Distance between co-located cameras C2 and C3 (GT ≈ 0 ; structural consistency test)

Note that ATE and RPE trans are in arbitrary units (scene scale is determined by the Sim(3) alignment, not metric ground truth).

6.2 Comprehensive Comparison

Table 2 presents the progressive improvement from raw Pi3X output through our full method.

6.3 Analysis

Several patterns are evident from Table 2:

Classical post-processing helps translation, not rotation. The pipeline (row 2) reduces ATE by 10% (pose graph with co-location constraints) but provides no rotation improvement. This confirms that the rotation error in Pi3X’s output is systematic bias, not statistical noise amenable to averaging.

Self-conditioning improves rotation at the cost of translation. Naïve conditioning (row 3) achieves a 20% ARE reduction but increases ATE by 3%. This asymmetry motivates the decoupled approach.

Hybrid fusion resolves the tradeoff. Row 4 shows that by taking rotations from the conditioned run and translations from the unconditioned run, we achieve rotation improvement (3.45°) while preserving—and slightly improving—translation (ATE = 0.159). At only 2× the compute cost, this is the most practical configuration.

MC iteration pushes rotation further. The full method (row 5) gains an additional 9% rotation improvement over single-step hybrid fusion ($3.45^\circ \rightarrow 3.13^\circ$), at the cost of $13\times$ total compute. This may be justified for high-precision applications but is not necessary for most practical deployments.

Overall improvement. Comparing row 1 (raw model output) to row 5 (full method): ARE improves by 28% ($4.37^\circ \rightarrow 3.13^\circ$), RPE rotation by 27%, and RPE translation by 21%, while ATE improves by 10%. The co-location distance drops from 0.088 to exactly 0.000 (hard constraint).

Oracle upper bound confirms the mechanism. The final row of Table 2 reports Pi3X conditioned on *ground truth* camera poses. This oracle achieves $\text{ATE} = 0.097$ and $\text{ARE} = 1.22^\circ$, representing 72% and 40% reductions over the unconditioned baseline, respectively. Two conclusions follow. First, pose conditioning is not merely a regulariser—it provides a genuine geometric prior that substantially improves Pi3X’s reconstruction. Second, the gap between our self-refined result ($\text{ARE} = 3.13^\circ$) and the oracle ($\text{ARE} = 1.22^\circ$) quantifies the remaining headroom: better pose priors (e.g., from a coarse SfM initialisation or a small number of known correspondences) would yield further gains without architectural changes. The monotonic improvement from unconditioned $\rightarrow$ self-conditioned $\rightarrow$ oracle rules out the possibility that our pipeline’s gains are coincidental, and demonstrates that the conditioning pathway is functioning as intended.

7 Ablation Studies

7.1 Why Hybrid Fusion is Necessary

Naïve iterative refinement (feeding full conditioned output back without decoupling) degrades translation while improving rotation:

Table 3: Naïve vs. hybrid iterative refinement over 5 iterations.

Method	ATE↓	ARE↓	ΔATE	ΔARE
Unconditioned baseline	0.161	4.39°	—	—
Naïve self-refine (iter 5)	0.195	3.60°	+21%	−18%
Hybrid MC (iter 2)	0.160	3.13°	−1%	−29%

The Sim(3) scale ambiguity in Pi3X outputs means that conditioned translations accumulate bias across iterations. The hybrid approach avoids this by anchoring translations to the unconditioned source.

7.2 Noise Robustness of Pose Conditioning

We test Pi3X’s conditioning with ground truth poses corrupted by varying levels of Gaussian noise: Key observations:

- Conditioning tolerates up to $\sim 5^\circ$ of rotation noise
- Beyond 10° , the conditioning signal becomes misleading and breaks the model
- Our self-conditioned prior ($\sim 4.4^\circ$ error) is at the edge of the useful regime, explaining the diminishing returns beyond 2–3 iterations

Table 4: Effect of conditioning pose quality on Pi3X output (after full pipeline). Translation noise is isotropic Gaussian in the scene coordinate system.

	Rot. noise ($^{\circ}$)	Trans. noise	ATE $\downarrow$	ARE $\downarrow$
0 (perfect GT)		0	0.097	1.22 $^{\circ}$
1		0.05	0.087	1.30 $^{\circ}$
2		0.10	0.080	1.60 $^{\circ}$
5		0.20	0.077	3.34 $^{\circ}$
10		0.50	0.368	180 $^{\circ}$
20		1.00	0.951	180 $^{\circ}$

7.3 Convergence Behavior

The iterated hybrid MC method converges within 2–3 iterations:

Table 5: Convergence of iterated hybrid MC refinement (4 samples per iteration).

Iteration	ATE	ARE ($^{\circ}$)	RPE rot ($^{\circ}$)	RPE trans
0 (unconditioned)	0.161	4.39	6.26	0.375
1	0.162	3.43	4.88	0.321
2	0.160	3.13	4.58	0.309
3	0.163	3.38	4.70	0.307
4	0.160	180 $^{\circ}$	4.78	8.55

Iteration 2 is the sweet spot. Beyond iteration 3, the conditioned rotations begin to drift and can cause orientation flips (iter 4). We recommend stopping after 2–3 iterations or when ARE improvement is $< 0.1^{\circ}$.

7.4 Sim(3) Constraint Does Not Help

We tested Sim(3)-aligning the conditioned output back to the conditioning frame before each iteration (to prevent scale drift). Results are identical to naïve iteration, confirming that translation degradation is not caused by scale drift but by systematic bias in how Pi3X uses pose conditioning:

Table 6: Sim(3)-constrained vs. naïve iteration (5 iterations).

Iter	Sim3 ATE	Sim3 ARE	Naïve ATE	Naïve ARE
0	0.152	4.69 $^{\circ}$	0.152	4.69 $^{\circ}$
1	0.160	3.50 $^{\circ}$	0.160	3.50 $^{\circ}$
2	0.180	3.31 $^{\circ}$	0.179	3.29 $^{\circ}$
5	0.197	3.61 $^{\circ}$	0.195	3.59 $^{\circ}$

8 Discussion

8.1 Practical Recommendations

1. **Always run the baseline pipeline** (Stage 1)—gauge fixing, outlier rejection, and pose graph optimization provide consistent improvements at negligible cost.
2. **One hybrid refinement step** (Stage 3) offers the best cost-benefit tradeoff: $2\times$ compute for $\sim 21\%$ rotation improvement.
3. **Full MC refinement** (Stage 4) is recommended when compute budget allows: $13\times$ compute for 28% rotation improvement.
4. **Never iterate naïvely**—always use hybrid fusion to protect translations from Sim(3) bias accumulation.
5. **Stop after 2–3 iterations**—monitor ARE convergence and terminate when improvement < 0.1 .

8.2 Limitations

- The method cannot break the $\sim 3^\circ$ ARE floor without external priors (the contraction rate $\lambda \approx 0.8$ means the fixed point is at $\sim 3^\circ$)
- Translation improvement is bounded by the unconditioned baseline’s accuracy
- The $13\times$ compute cost of full MC refinement may not justify the marginal gain over a single hybrid step for latency-sensitive applications
- Convergence is empirical, not guaranteed—iteration 4+ can diverge due to orientation flips
- The method assumes access to intrinsic camera parameters (available from COLMAP or manufacturer specs)

8.3 Future Directions

- **Depth conditioning:** Pi3X also accepts depth maps as input. Monocular depth priors (e.g., Depth Anything v2) could provide complementary geometric signal without requiring ground truth poses.
- **Partial conditioning:** Condition on a subset of cameras with higher confidence, letting the model propagate information to uncertain cameras.
- **Adaptive noise scheduling:** Learn the optimal perturbation magnitude per iteration rather than using a fixed $\sigma_R = 2$.
- **Learned fusion weights:** Replace the $\max(w^{\text{cond}}, w^{\text{uncond}})$ heuristic with learned per-edge reliability.
- **Breaking the 3° floor:** Incorporate sparse LiDAR points or a pre-calibrated camera pair to provide a stronger anchor for the conditioning signal.

9 Conclusion

We have shown that self-conditioning in learned geometry models exhibits a fundamental asymmetry: rotation contracts toward a fixed point on $SO(3)$ while translation diverges due to $Sim(3)$ scale ambiguity. Exploiting this asymmetry via decoupled hybrid fusion—sourcing rotation from conditioned inference and translation from unconditioned inference—yields consistent improvements without the degradation of naïve iteration. Combined with Monte Carlo perturbation, our method achieves 28% rotation improvement over the raw Pi3X baseline on a challenging wide-baseline, feature-adversarial calibration task, with no external calibration data required.

The contraction rate ($\lambda \approx 0.8$) and resulting accuracy floor ($\sim 3^\circ$) are properties of the model and scene, not the method. Breaking this floor requires either stronger external priors or architectural changes to the conditioning mechanism. We believe the decoupled self-refinement framework generalises to any learned geometry model with a pose conditioning interface, and the contraction analysis provides a principled tool for predicting when iterative refinement will help.

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